



UCWS SINGAPORE 2026 · SKILLS TRACK

BakerySense

A self-evolving operations harness **for bakeries and perishable SMEs.**

It turns sales history into tomorrow's bake plan — then learns from its own decisions, evolving a different playbook for every shop, under one-tap human approval.

bakerysense.swmengappdev.workers.dev · github.com/Whyme-Labs/bakerysense

THE PROBLEM

30-40%

of everything an independent bakery bakes is thrown out.

The waste isn't ignorance — it's uncertainty.

Forecasts are never perfectly calibrated. Every branch behaves differently. No busy owner notices that croissants quietly run 25% high every Wednesday at one shop, and every Sunday at another. That blind spot is pure margin on the floor.

WHAT IT DOES

Tomorrow's bake plan, as a decision — not a chatbot

BakerySense

Dashboard

Chat

Display case

Admin

Bukit Bintang

no connector

Settings

Sign out

TENANT - DEMO

Today's bake plan

Close out today

Ask Gemma for a narrative →

Branch brn_fxjy1wpy7y6 · 2026-05-29

THREE PLANS PER SKU

Pick a plan to commit. Numbers are expected outcomes from the quantile forecast under your cost ratio.

CROISSANT

CONSERVATIVE

75

Expected waste2 units

Stockout chance75%

Expected sold73 units

Commit

BALANCED

70

Expected waste1 units

Stockout chance82%

Expected sold69 units

Commit

AGGRESSIVE

121

Expected waste28 units

Stockout chance30%

Expected sold93 units

Commit

PAIN AU CHOCOLAT

CONSERVATIVE

70

Expected waste3 units

Stockout chance70%

Expected sold67 units

Commit

BALANCED

60

Expected waste1 units

Stockout chance83%

Expected sold59 units

Commit

AGGRESSIVE

121

Expected waste32 units

Stockout chance30%

Expected sold89 units

Commit

TRADITIONAL BAGUETTE

CONSERVATIVE

108

Expected waste4 units

Stockout chance81%

Expected sold104 units

Commit

BALANCED

140

Expected waste19 units

Stockout chance20%

Expected sold121 units

Commit

AGGRESSIVE

121

Expected waste8 units

Stockout chance64%

Expected sold113 units

Commit

Three plans per SKU.
Conservative · Balanced · Aggressive —
each with expected waste, stockout
chance, and units sold, picked by
newsvendor math from a quantile
forecast under the shop's own cost
ratio. One tap commits the call.

It learns from its own decisions

1

diagnose

Classify every miss.
Stockout-censored,
operator override, one-off
event, or a genuine skill
fault — only the last one is
learnable.

2

propose

Emit a bounded, learning-
rate-capped correction to a
version-controlled skill
artifact (not the model).

3

validate

Re-score on a strictly held-
out window. The edit must
measurably cut error before
anyone sees it.

4

approve

The owner approves in one
tap. Controlled autonomy
— every edit is validated
and auditable.

The forecasting model stays frozen. The skills evolve.

SELF-INSPECTION IN ACTION

Corrections it taught itself, waiting for review

BakerySenseDashboardChatDisplay caseAdmin

Bukit Bintangno connectorSettingsSign out

ConnectorsDataUsersBranchesModelHarnessAudit

TENANT · DEMO

Harness evolution

Corrections the harness has learned from this tenant's own sales, each validated on a held-out window. Approving an item activates a new skill version; the next forecast applies it.

RUN INSPECTION

Replay this branch's recent traces, diagnose misses, and propose a validated correction.

Bukit Bintang

Run inspection

forecast · Subang Jaya

WAPE 3.4% → 2.2%

CROISSANT on Sun: forecast over by ~23% across 8 days → lower multiplier 1 → 0.8

CROISSANT on Sun: 1 → 0.8

Approve

Reject

forecast · Bukit Bintang

WAPE 3.0% → 2.0%

CROISSANT on Wed: forecast over by ~20% across 8 days → lower multiplier 1 → 0.8

CROISSANT on Wed: 1 → 0.8

Approve

Reject

Every night, the harness replays its traces and proposes validated fixes.

Bukit Bintang: croissants +25% on Wednesdays.

Held-out WAPE 3.0% → 2.0%.

CROISSANT · Wed 1.00 → 0.80

Same brand, same model — different playbooks

Bukit Bintang

forecast · Bukit Bintang

WAPE 3.0% → 2.0%

CROISSANT on Wed: forecast over by ~20% across 8 days → lower multiplier 1 → 0.8

CROISSANT on Wed: 1 → 0.8

Approve

Reject

CROISSANT · Wed → 0.80

learned Wednesdays on its own

Subang Jaya

forecast · Subang Jaya

WAPE 3.4% → 2.2%

CROISSANT on Sun: forecast over by ~23% across 8 days → lower multiplier 1 → 0.8

CROISSANT on Sun: 1 → 0.8

Approve

Reject

CROISSANT · Sun → 0.80

learned Sundays on its own

Each shop becomes its own agent with its own skill set — branch skills are sparse overrides on brand defaults. A federation of self-evolving shops.

EXECUTION OVER POLISH

A working loop on live infrastructure — not a mockup

285

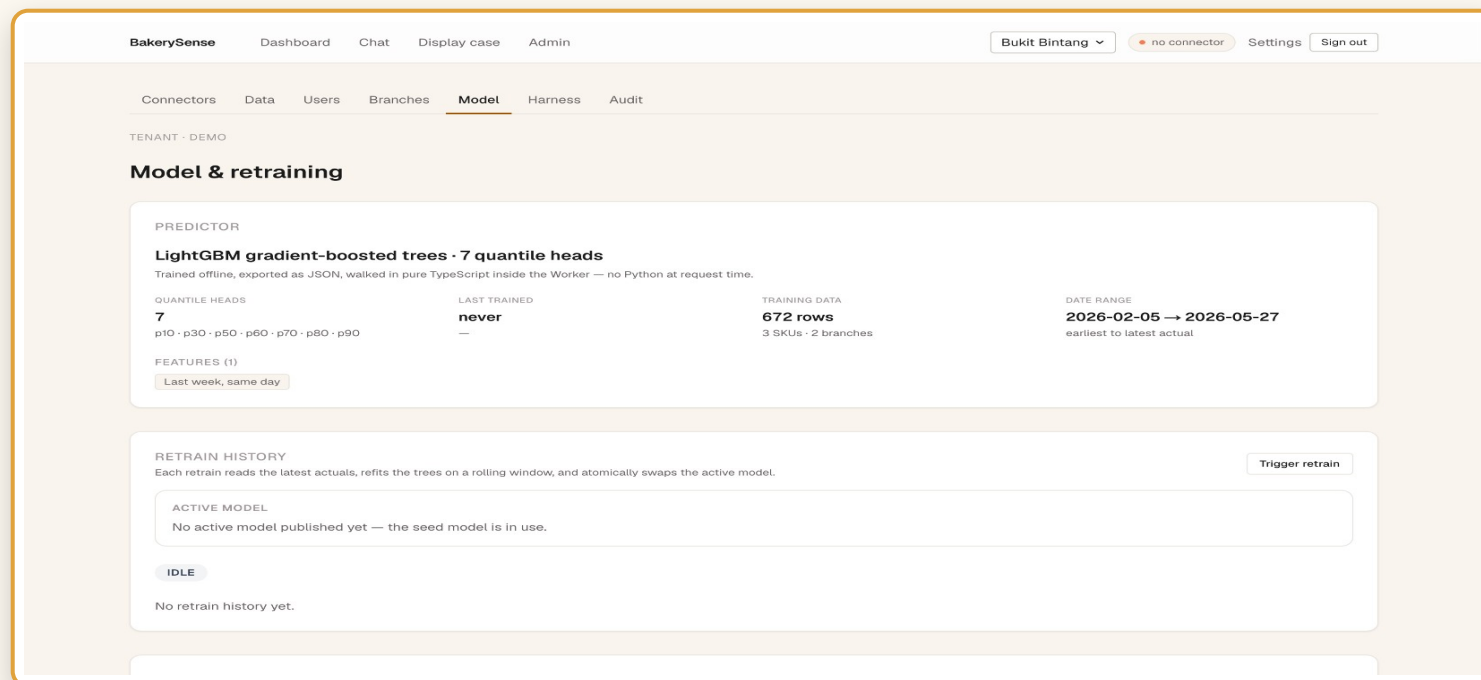
tests passing

100%

deterministic numbers

Live

on Cloudflare D1



Deterministic numbers, semantic Gemma.

Quantities come from a LightGBM quantile forecaster + newsvendor — never hallucinated. Gemma 4 is the narration and explanation layer. That split is what makes recommendations trustworthy to a real operator — and it's what gets the harness's edits past validation instead of vibes.

GLOBAL SCALABILITY

One harness, every perishable business

● Bakeries

● Cafés

● Grocers

● Florists

● Cloud kitchens

● Central kitchens

The brand→branch skill hierarchy scales to chains. Only **learned rule diffs** propagate — not raw sales — so it preserves each shop's data privacy while sharing what works.

Real product value · execution quality · global scalability.



GET STARTED

Try it in 30 seconds

Live demo `bakerysense.swmengappdev.workers.dev`

Sign in `demo@bs.co` / `Password2026Password` (slug: demo)

Money shot Admin → Harness – two branches, two different learned fixes, awaiting approval.

Repo

`github.com/Whyme-Labs/bakerysense`